Moscode Formalisation of Dijkstra Algorithm

Primitives : relax function

1. relax function

$$(\text{update } d\ U\ v)(w) = \begin{cases} \min\{d[w], d[v] + wt(u,w)\} & \text{if } w \in U \\ d[w] & \text{else} \end{cases}$$

It is parametrised on given Graph $G = (V, E, wt)$. Takes in distance mapping d, set of unvisited vertices U, current vertex v and returns a new distance mapping

2. Graph ADT : A undirected weighted graph is given by 3 tuple (V, E, wt) with vector of Naturals V being set of vertices, $E \subseteq V \times V$ being set of edges and $wt : E \times E \mapsto \mathbb{N}$

3. Natural Numbers, Ordering : $\mathbb{N}, \leq, \min$, $\forall n \in \mathbb{N} : n < \infty$

4. Operations on Sets/Maps : Union, Difference, min, argmin

Definitions

$$I = \{\text{Set of Graphs}\} \times \mathbb{N}$$

$$X = 2^{\mathbb{N}} \times (\mathbb{N} \mapsto \mathbb{N} \cup \{\infty\})$$

$$A = \mathbb{N} \mapsto \mathbb{N}$$

$$\rho((V, E, wt), s) = (V, \lambda n.\ 0 \text{ if } n = s \text{ else } \infty)$$

$$F(U, d) \begin{cases} (U, d) & \text{if } U = \{\} \\ (U - \{w\}, \text{update } (d)(U - \{w\})(w)) & \end{cases}$$

where $w = \underset{v \in U}{\text{argmin}}\ d(v)$

$$\Omega(U, d) = d$$

Invariant and Bound Functions, along with Proofs

Let $\sigma(u, d) := |u|$

σ is well founded proof:

$\forall x \in X$ such that $x \neq Fx$

let $x = (u, d)$, $Fx = (u', d')$

$u \neq \{\}$. $\because$ If $u = \{\}$ then $Fx = x$ contradiction

So $u' = u - \underset{a \in u}{\text{argmin}}\ da$

$\sigma(Fx) = |u'| = |u| - 1 < |u| = \sigma(x)$

$\therefore \sigma(Fx) < \sigma(x)$ ∎

Let $\phi(x) := \Omega \circ F^{\infty}(x)$ $\qquad F^{\infty}x := x$ if $x = Fx$ else $F^{\infty}x$

Since σ is well founded, fixed point exists

$\forall x \in X,\ \theta(Fx) = \theta(x)$

proof For any $x \in X$,

$\theta(x) = \Omega \circ F^{\infty}(x) = \Omega \circ F^{\infty} \circ F(x) = \Omega \circ F^{\infty}(Fx) = \theta(Fx)$

$\forall x \in X,\ x = Fx \Rightarrow \theta x = \Omega x$

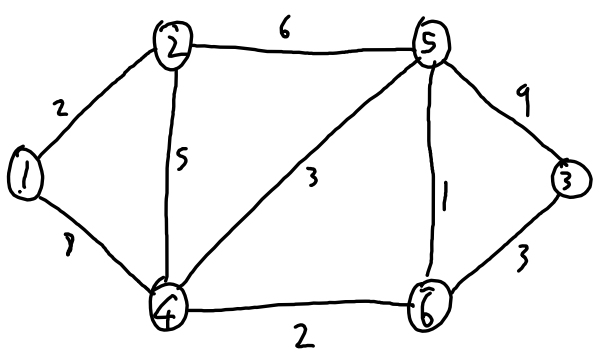
proof For all $x \in X$, s.t $x = Fx$

$\theta x = \Omega \circ F^{\infty}(x) = \Omega(F^{\infty}(x)) = \Omega(x)$

$\forall i \in I,\ \theta(\rho i) = \Omega \circ F^{\infty} \circ \rho(i)$

proof $\theta(\rho i) = \Omega \circ F^{\infty}(\rho i) = \Omega \circ F^{\infty} \circ \rho(i)$

Trial Run



$i = ((V, E, wt), s)$ where

$V = \{1 \ldots 6\}$

$E = \{(1,2), (1,4), \ldots, (5,3), (6,3)\}$

$wt(1,2) = 2$

$wt(1,4) = 8$

$\vdots$

$wt(5,3) = 1$

$wt(6,3) = 3$

$s = 1$

	u	d	σ
$\rho(i) = x_0$	$\{1,2,3,4,5,6\}$	$[0,\infty,\infty,\infty,\infty,\infty]$	6
Fx_0	$\{2,3,4,5,6\}$	$[0,2,\infty,8,\infty,\infty]$	5
F^2x_0	$\{3,4,5,6\}$	$[0,2,\infty,7,8,\infty]$	4
F^3x_0	$\{3,5,6\}$	$[0,2,\infty,7,8,9]$	3
F^4x_0	$\{3,6\}$	$[0,2,17,7,8,9]$	2
F^5x_0	$\{3\}$	$[0,2,12,7,8,9]$	1
F^6x_0	$\{\}$	$[0,2,12,7,8,9]$	0
$x_f = F^{\infty}x_0$	$\{\}$	$[0,2,12,7,8,9]$	0

$\Omega(x_f)$

$$\Omega \circ F^{\infty} \circ \rho(i) = \begin{cases} 1 \to 0 \\ 2 \to 2 \\ 3 \to 12 \\ 4 \to 7 \\ 5 \to 8 \\ 6 \to 9 \end{cases}$$